Community Hospital System Application Specification

Global Context: Globalization And Sustainability

MYP Community Project 2024-2025

Name: Qiwen He (9.3 Markus), Junbo Zhang (9.3 Tom), Runze Jiang (9.3 Russell)

Grade: 9

Supervisor: Xiaohui Wang (Ms. Monica)

School Name: Beijing Huijia Private School

Word Count: 1293

Table of Contents

Community Hospital System Application Specification 1
Table of Contents 2
Introduction 3
Proximate Community Hospital Geographic Information System 4
- **Functionality** 4
- **Information Components** 4

User Health Profile System 5
- **Functionality** 5
- **Information Components** 5

Consultation Data Record System 7
- **Functionality** 7
- **Information Components** 7

Online Reservation System 8
- **Functionality** 8
- **Information Components** 8

Online Consultation System 9
- **Functionality** 9

User Interface System 10
- **Functionality** 10

Introduction

The Community Hospital System Application is an application dedicated to the enhancement on the experience of both community hospitals and visitors to the hospitals. The community hospital can improve management of data and reservation, gain online visitors, while users can gather more information about community hospitals and contribute to efficient selection of hospital visits.

Proximate Community Hospital Geographic Information System

Functionality

The design of this system primarily involves an integrated information system of community hospital, including geographical information system and hospital information.

The geographical information system collects the exact location and addresses of community hospitals in the range of Beijing municipal area. The data will be selectively shown to the user according to the collection of user's GPS location. Particularly, community hospitals in the range of ten kilometers will appear on a map included in the user interface with basic information, including exact distance, type of treatment (Chinese Medicine/Western Medicine) and availability of online consultation.

The system should also classify the specific type of community hospital and provide a filter option for the user, this increases efficiency for user to select optimal hospitals.

The form of information system should be dynamic in integrating with household devices, including watches, phones, computers for convenient check of information.

Information Components

The information of community hospitals will be collected through three ways:

1. Direct input from developer for community hospitals with complete information findable online.
2. Application from community hospitals by providing proof, and then be approved by developer.
3. User submission of community hospital real site proof, and then be approved by developer.

The above methods also apply as ways to update information of community hospitals.

User Health Profile System

Functionality

The user health profile system features input and storage function. Community hospital can input user health profile and store it into the storage data system. User can input parts of daily data for clinical uses. The data is updated to online servers, and thus can connect with the user side in the form of application, the user side can monitor the validity of the data from community hospital.

Data will be given different tags, some tags are equipped with automatic reminders set, for instance a regular reminder of low-sugar consumption for users under the tag diabetes. The reminder can also be sent when the user-side data reaches a specific point, for instance excess sugar consumption recorded in self-input.

User health profile system is privacy related. Thus, data protection is included. Community hospital shall have a password-secured account for data accessing and input. User side shall also have a password-secured account for data access and reminders. Static data shall also be encrypted, for instance in AES-256, transport layer security is needed for communication of data. The information security should comply with Personal Information Protection Law, or PIPL of China. A data backup shall also be included.

Information Components

The information included in this system mainly involves two sections, which is hospital-side information and user-side information.

The hospital-side input mainly involves these sections:

1. Name (M)
2. Age and Gender (M)
3. Contact Information (M)
4. Health Insurance Information (M)
5. Chief Complaint(s) (M)

6. Visits to the hospital(M)
7. Ongoing Diagnosis

*M stands for mandatory input.

The user-side input mainly involves these sections:

1. Weight
2. Blood Pressure
3. Blood Sugar Levels
4. Dietary Diet according to nutrition intake
5. Health Goals
6. Sleep Status (Primarily Insomnia)
7. Smoke/Alcohol intake

Consultation Data Record System

Functionality

Consultation data record system is strongly related to the user health profile system. It is a detailed profile of the users' consultations in a specific hospital. The data will be manually inputted by the community hospital side.

The data can be linked with the same person recorded in the user health profile. The detailed consultation will be transferred to the user side as local storage; thus, the user can access the records offline and conveniently.

The data should also be properly secured under the PIPL of China, security mainly involves in encryption, creating passworded account of community hospital and user, and transport layer security.

Information Components

The consultation data mainly involves separate consultation record with such included:

1. Name and contact information (For filtering, not included in user side)
2. Chief Complaint
3. Investigation Type
4. Clinical Findings
5. Primary and Secondary diagnosis (ICD-10 Codes, Optional)
6. Treatment plan
    a. Medication Prescribed
    b. Instructions
    c. Referral
7. Doctor Notes
8. Scheduled Further Appointments

User does not have the right to modify the above data but can consult for errors.

Online Reservation System

Functionality

The online reservation system primarily serves as a method for patients to secure a time for consultation considering that community hospitals are public resources. Community hospital shall select and display available dates and periods. The number of patients reservable in a period will be determined by the hospital, while the users can manually select suitable time periods.

Optionally, the user can input chief complaint online to increase efficiency for their consultation. Reservation data will be automatically recorded and stored both in the community hospital and user account.

Reservations are subject to changes. Therefore, notifications are crucial when changes occur. Instant notification shall be sent to the user after a change of availability of a reservation. The availability of distant dates is usually unanticipated; thus, users can follow preferred community hospital and regular receive notifications of availability changes.

Information Components

The information featured in the system mainly involves three sections, which is community hospital availability, user reservation profile and reservation record.

Community hospital availability information is inputted by the hospital and updates to the server, it includes available date and time period per fifteen minutes. The availability of a specific time period can be changed after reaching a limit set by the hospital (i.e. 2 people) or manually updated.

User reservation profile and record includes such detail:

1. Name and Contact Information
2. Reserved Time
3. Complaint (Optional)

Online Consultation System

Functionality

The online consultation system is an optional method for community hospital to receive online patients. Online consultation must comply with the Chinese “Management Measures for Online Diagnosis and Treatment Law”.

Patients with proof or with existing record of community hospital visits can apply for online consultation with same complaints compared to in-person visits. The system includes an online text communication tool which can be used for the consultation. History of online consultation will be archived, and user will be able to find previous consultations in the application

Online consultation will also be recorded in the user health profile. The data security involves in both the storage of consultation record and the message communication. Message will be end-to-end encrypted for safety; the general data security will follow the PIPL of China.

User Interface System

Functionality

User interface system is an integration of functions of previously mentioned systems. It involves two main sections, the community hospital section and user section.

The user section mainly features health profile, online consultation, online reservation, community hospital map, consultation history and reservation history. The community hospital section mainly features health profile input, online consultation, reservation availability input and hospital information input.